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Adaptive parallel filter method for active cancellation of road noise inside vehicles

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ABSTRACT

Active noise cancellation (ANC) is an efficient and effective method for suppressing low frequency disturbances generated by tyre–road interaction in a vehicle. There are significant differences in road noise inside the vehicle under different driving conditions. However, fixed-filter ANC strategies are unable to achieve superior noise attenuation for different types of noise, and conventional adaptive algorithms require prolonged adaptation which is detrimental to dynamic noise reduction. Thus, this study proposes an adaptive parallel filter method (APFM) that quickly responds to and suppresses varying vehicle interior road noises. The APFM is divided into two phases: the tuning phase and control phase. In the tuning phase, a set of potential noises is adopted to train the corresponding optimal fixed filters. During the control phase, the most appropriate parallel control filter is first selected for each frame based on a selection mechanism to minimise the energy of the estimated error signal. Then, a delayless variable step size normalised frequency-domain block adaptive filter algorithm is adopted to update the selected control filter. In addition, the computational complexity of the proposed APFM is analysed. Numerous simulations and real time in-vehicle experiments conducted using multi-channel ANC headrests have confirmed the feasibility and effectiveness of the proposed algorithm. It is evident that the APFM achieves fast noise reduction response and low steady-state error in attenuating varying in-vehicle road noise, and retains superior performance in the presence of non-pre-trained disturbances.

1. Introduction

To improve passenger ride comfort and mental health, active cancellation of road noise inside vehicles has been studied for more than 20 years [1–5]. Road noise, which typically falls within the frequency range of 50–500 Hz, is transmitted into the passenger compartment through vibrations generated by the interaction of tyres and road via suspension and bodywork [6,7]. Active noise cancellation (ANC), based on the principle of acoustic interference, can effectively suppress low frequency disturbances with low volume and weight [8–10]. Therefore, ANC headrests are used to attenuate vehicle interior road noise and create a local zone of quiet around the occupants' head [11–14].

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Feedforward ANC structures have been developed to reduce broadband road noise using reference signals obtained by directly measuring of vibrations with accelerometers [3,15]. The position of the reference sensors can be determined by coherence analysis, principal component analysis, and the effective independence method using the Fisher information matrix [16]. To achieve a reasonable level of noise reduction (NR), at least six accelerometers are required to obtain reference signals that are highly correlated with the error signals [3]. At the same time, ANC systems must achieve instantaneous NR and maintain constant stability [17]. Therefore, such multi-channel ANC systems impose high requirements on the convergence speed, steady-state error, and computational complexity of the adaptive filter algorithms [6,18,19]. The filtered-x least mean square (FxLMS) algorithm is the most commonly used adaptive algorithm in ANC due to its effectiveness and simplicity [20,21]. However, the convergence performance of the FxLMS algorithm is susceptible to the eigenvalue spread for multiple reference signals [6,20]. And the control filter is updated point-by-point, resulting in high computational complexity [20]. Therefore, normalised frequency-domain block FxLMS (NFBFxLMS) algorithms are adopted to improve the convergence speed and reduce the computational cost [22–25]. In addition, to overcome the trade-off between convergence speed and steady-state performance due to fixed step size, various variable step size (VSS) strategies have been developed for adaptive filter algorithms [26–28]. The objective of VSS methods is to adaptively adjust the step size to a larger value for fast convergence speed and then to a smaller value for low steady-state error [26]. These adaptive algorithms must focus on a balance between NR performance and stability [29]. In comparison, fixed-filter ANC methods, represented by the Wiener filter algorithm, avoid tedious adaptation processes and potential instability by off-line design with fixed control filter coefficients [30]. However, fixed-filter ANC methods are pre-trained for a specific noise type, leading to a significant degradation of the NR performance for other types of noise [31].

To achieve a fast and robust response to various types of noise, Shi et al. [32] proposed a selective ANC (SANC) method. The controller used was selected from various pre-trained fixed filters based on frequency-band-matching. To speed up the selection, Wen et al. [33] used empirical wavelet transform to quickly extract signal features and select a suitable controller for SANC. Deep learning methods have also been applied to the SANC method to automatically learn relevant features from noisy datasets and select the optimal fixed filter for different types of noise using a co-processor [31,34]. However, the NR performance of these SANC methods deteriorates significantly when dealing with non-pre-trained noises or primary path variations [31]. Luo et al. [35] combined deep learning with an adaptive filter algorithm to address this problem. A one-dimensional convolutional neural network was used to select the most appropriate pre-trained fixed controller for each frame, and then the fixed filter coefficients were updated using the normalised FxLMS algorithm at the sampling rate [35]. However, for multi-channel systems, the process of training models with deep learning is complex and the co-processor requirements are high [36–38]. The slow convergence of the normalised FxLMS algorithm is detrimental to the reduction of time-varying disturbances [39]. The above SANC-based methods hardly introduce or analyse the case of multiple reference signals and lack the corresponding real time experimental validation. Especially in multi-channel systems, the computational complexity of the feature extraction methods for SANC will increase significantly, while the accuracy may decrease [40,41]. Therefore, these SANC-based methods are probably inappropriate for active road noise cancellation inside vehicles at present and are not further compared and discussed in the rest of this study [42].

Based on the above discussions, we propose an adaptive parallel filter method (APFM) that can quickly respond to and suppress varying vehicle interior road noise and is adopted in multi-channel feedforward ANC headrest systems. The APFM is divided into two phases: the tuning phase and control phase. In the tuning phase, a series of potential primary noises were used to train the corresponding optimal fixed control filters [34,35,43]. During the control phase, the most appropriate parallel control filter is selected based on a selection mechanism to minimise the energy of the estimated error signal within a frame [44]. Then, a delayless VSS normalised frequency-domain block FxLMS (VSS-NFBFxLMS) algorithm is proposed to update the selected control filter at the frame rate. In this way, APFM achieves fast NR response and low steady-state error as a robust method for suppressing varying noise and achieves superior performance in attenuating non-pre-trained disturbances.

The remainder of this paper is organised as follows. Section 2 presents the analytical derivation and computational complexity analysis of the proposed APFM. Section 3 introduces numerical simulations performed using the transfer functions measured from an ANC headrest system. Section 4 presents the real time experimental results conducted on an electric vehicle and the future lines of investigation. Finally, Section 5 concludes the study.

2. Proposed APFM

The proposed APFM is divided into the tuning and control phases. During the training phase, it is necessary to obtain a series of optimal fixed control filters corresponding to all potential primary noises using either the adaptive filter algorithm or the Wiener filter algorithm [34]. In the control phase, the selection mechanism is adopted to find a pre-trained control filter that is best suited for the current primary noise [44]. The VSS-NFBFxLMS algorithm is then used to update the selected filter to account for non-pre-trained disturbances and varying primary paths. In this section, a multi-channel ANC system is constructed with I reference sensors, J secondary loudspeakers, and M error microphones. Q types of potential primary noise are used to obtain the corresponding control filters during the tuning phase. Multi-channel time-domain signal vectors are presented in lower-case bold, and the matrices of the corresponding frequency responses are shown in upper-case bold. The superscript $\hat{}$ denotes the estimated signal and modelled transfer function.

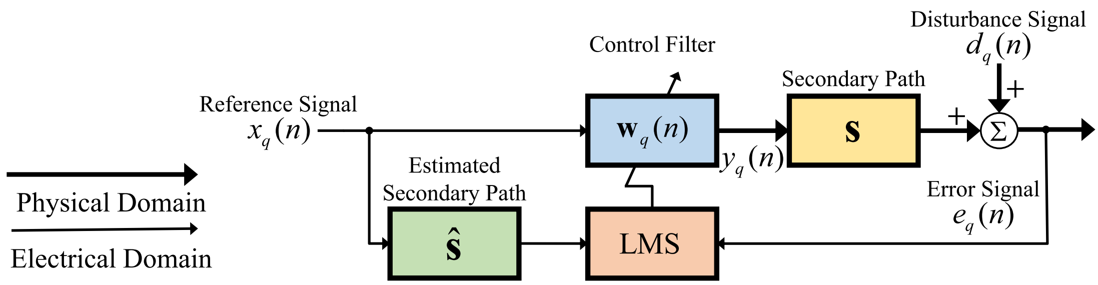

Fig. 1. Block diagram of the tuning phase of the proposed APFM.

2.1. Tuning phase

In the tuning phase, the FxLMS algorithm can be used for all potential noises to train the corresponding fixed control filters, as shown in Fig. 1. In the qth noise environment, the jth output signal at time sample n is denoted as

$$y_{q,j}(n) = \sum_{i=1}^{I} \mathbf{w}_{q,ji}^{T}(n)\mathbf{x}_{q,i}(n), \tag{1}$$

where $\mathbf{x}_{q,i}(n) = \left[x_{q,i}(n), \dots, x_{q,i}(n-L+1)\right]^T$ denotes the ith reference signal vector with the last L samples, $\mathbf{w}_{q,ji}(n) = \left[w_{q,ji}(n), \dots, w_{q,ji}(n-L+1)\right]^T$ represents the control filter vector between the ith reference signal and the jth output signal, and the superscript $(\cdot)^T$ denotes the transpose operator. To minimise the energy of the M error signals, the cost function is defined as

$$J_q(n) = \sum_{m=1}^{M} \mathbb{E}\left\{e_{q,m}^2(n)\right\} = \sum_{m=1}^{M} \mathbb{E}\left\{\left[d_{q,m}(n) + \mathbf{s}_{mj}^T \mathbf{y}_{q,j}(n)\right]^2\right\}, \tag{2}$$

where $\mathbb{E}\{\cdot\}$ denotes the expectation operator; $d_{q,m}(n)$ and $e_{q,m}(n)$ represent the disturbance signal and error signal at the mth error microphone, respectively; $\mathbf{s}_{mj}$ indicates the impulse response of the secondary path, which is from the jth secondary source to the mth error microphone; and $\mathbf{y}_{q,j}(n) = \left[y_{q,j}(n), \dots, y_{q,j}(n-G+1)\right]^T$ is the output signal vector with a length of G taps. Based on the FxLMS algorithm [20], the control filter is obtained, which is denoted as

$$\mathbf{w}_{q,ji}(n+1) = \mathbf{w}_{q,ji}(n) - \tilde{\mu} \sum_{m=1}^{M} \mathbf{r}_{q,mji}(n) e_{q,m}(n), \tag{3}$$

where $\tilde{\mu}$ denotes the step size, $\mathbf{r}_{q,mji}(n) = \left[r_{q,mji}(n) \dots, r_{q,mji}(n-L+1)\right]^T$ is the filtered-reference signal vector, and $r_{q,mji}(n)$ is given by

$$r_{q,mji}(n) = \hat{\mathbf{s}}_{mj}^T \mathbf{x}_{q,i}(n), \tag{4}$$

where $\hat{\mathbf{s}}_{mj}$ indicates the estimated secondary path with length of G taps. After the controller converges to the optimal solution, the pre-trained fixed control filter $\mathbf{w}_q$ for the qth type of noise is obtained [31]. Then, the primary noise is switched to the next type to train the corresponding fixed controller and populate the control filter database [34]. The tuning phase can be conducted either by numerical simulations or real time experiments [34].

2.2. Control phase

Fig. 2 presents the diagram of the control phase of the APFM. During the control phase, the most appropriate control filter is first matched from parallel control filters at the frame rate based on the selection mechanism. For efficient suppression non-pre-trained primary noise, a delayless VSS-NFBFxLMS algorithm is then adopted to update the selected control filter at the frame rate. For pre-trained Q separate kinds of primary noise, there are $Q+2$ types of control filters to be selected. The first Q pre-filters are fixed control filters obtained by training the Q potential noises, respectively. The main difference between the two remaining control filters is that the $(Q+1)$th filter is the coefficient of the control filter for the current frame, while the $(Q+2)$th filter is a zero vector of $1 \times L$.

2.2.1. Selection mechanism

The selection mechanism is proposed and applied by us to improve the robustness of the virtual sensing method for attenuating varying noise [44]. The main idea of the selection mechanism is to estimate the disturbance signal using the internal model control and then calculate the corresponding parallel error signals for applying different parallel filters [10,44]. To reduce the potential

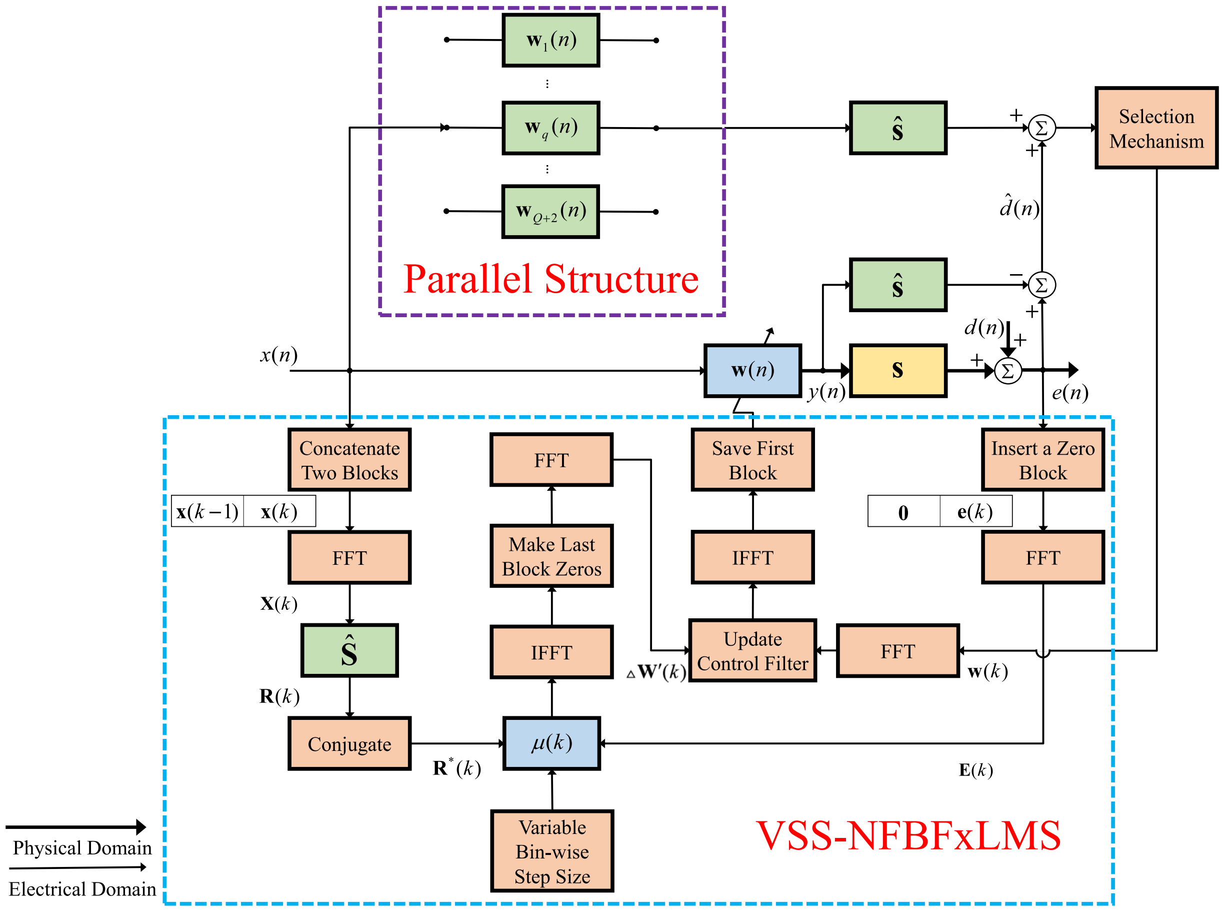

Fig. 2. Block diagram of the control phase of the proposed APFM.

influence of estimation errors and complexity cost, the most appropriate filter is selected frame by frame. Each frame yields a matched filter that minimises the energy of the estimated error signal and is used in the subsequent frame. Following this mechanism, we improve it and apply it to the APFM.

For the proposed APFM, the selection mechanism is improved in two ways. First, the selected control filter is used as the initial value for the adaptive filter algorithm to achieve a fast NR of non-pre-trained primary noises. The updated coefficients of the controller are added to the control filter database as the $(Q+1)$th controller. Another improvement is the addition of a filter with the coefficient set to $1 \times L$ zero vector as the $(Q+2)$th controller to improve the robustness of the APFM. This is because when the primary noise differs significantly in its spectral characteristics or primary path compared to the trained noises, the pre-trained control filters are susceptible to ineffectiveness and may cause the control filter to scatter [43].

Table 1 lists the details of the APFM, and the total number of multiplications and additions required per iteration for the APFM. The related source code is uploaded to the GitHub site at https://github.com/ucaszzq/Mat_APFM. Compared to the FxLMS and NFBFxLMS algorithms, the APFM requires more multiplications and additions, which are used in selecting the parallel filters and adjusting the step size [20,24]. It leads to the type of pre-trained fixed filters Q requiring careful determination based on the details of the application. However, the computational cost is well worth it to achieve faster NR response and lower steady-state error. The increased computational cost of the APFM is acceptable and can be compensated by a significant improvement in the attenuation performance of the varying broadband noise [19,27]. Thus, the APFM can be considered a promising algorithm for various ANC applications.

Referring to the storage requirement of the pre-filter database, the memory usage of the APFM includes two main parts. The first part is composed of the parallel control filter and the second part consists of the corresponding filter index. The storage consumption of the parallel control filter for the APFM is $IJL(Q+2)$ single-precision floating-point numbers. And the storage consumption for the filter index part is $(Q+2)$ integer numbers.

Table 1
Proposed APFM.

Input: Reference signals $x_i(n)$, error signals $e_m(n)$	**Output**: Output signals $y_j(n)$	
Initialisation: $\mathbf{w}_{q,ji}$, $\mathbf{w}_{ji}(0)=\mathbf{0}$, $\hat{\mathbf{s}}_{mj}$, $sum_q=0$	**Multiplications**	**Additions**
for $n=1,2,\ldots$		
1. Update the vectors $\mathbf{x}_i(n)$		
2. $y_j(n)=\sum_{i=1}^{I}\mathbf{w}_{ji}^T(n-1)\mathbf{x}_i(n)$	IJL	$IJ(L-1)$
3. Update the vectors $\mathbf{y}_j(n)$		
4. $\hat{d}_m(n)=e_m(n)-\sum_{j}^{J}\hat{\mathbf{s}}_{mj}^T\mathbf{y}_j(n)$	JMG	$JM(G-1)+M$
5. $r_{mji}(n)=\hat{\mathbf{s}}_{mj}^T\mathbf{x}_i(n)$	$IJMG$	$IJM(G-1)$
6. Update the vectors $\mathbf{r}_{mji}(n)$		
7. **for** $q=1,2,\ldots,Q+2$ (Parallel filter)		
8. $y_{q,j}(n)=\sum_{i=1}^{I}\mathbf{w}_{q,ji}^T\mathbf{x}_i(n)$, update the vectors $\mathbf{y}_{q,j}(n)$	$IJL(Q+1)$	$IJ(L-1)(Q+1)$
9. $\hat{e}_{q,m}(n)=\hat{d}_m(n)+\sum_{j}^{J}\hat{\mathbf{s}}_{mj}^T\mathbf{y}_{q,j}(n)$	$JMG(Q+1)$	$JM(G-1)(Q+1)+M(Q+1)$
10. $sum_q=sum_q+\sum_{m=1}^{M}\hat{e}_{q,m}^2(n)$	$M(Q+2)$	$(M-1)(Q+2)$
11. **end for**		
12. **if** $\mathrm{mod}(n,\mathrm{B})==0$ (Selection mechanism)		
13. Update the blocks $\mathbf{x}_i(k)$, $\mathbf{e}_m(k)$, and $\mathbf{r}_{mji}(k)$		
14. find h, s.t. $sum_h\le\min_{1\le q\le Q+2}sum_q$		
15. $\mathbf{w}_{ji}(n)=\mathbf{w}_{h,ji}(n)$		
16. Update $\mathbf{w}_{ji}(n)$ using the VSS-NFBFxLMS algorithm	$2B(IJM+3IJ+I+M)\log_2 2B$ $+B(6IJM+3I+M)+4I+2M$	$2B(IJM+3IJ+I+M)\log_2 2B$ $+B(2IJM+3I+M+2)+I(JL+2M)$
17. Update the $(Q+1)$th control filter: $\mathbf{w}_{Q+1,ji}(n)=\mathbf{w}_{ji}(n)$		
18. **end if**		
end for		

2.2.2. Delayless VSS-NFBFxLMS algorithm

The delayless VSS-NFBFxLMS algorithm saves computational effort compared with the FxLMS algorithm and ensures that the output signals are generated at the sampling rate to avoid latency [22,24]. The frequency-domain adaptive algorithm adopted is based on the overlap-save method and a fast Fourier transform to reduce the computational cost [45]. A combination with a VSS strategy further improves the convergence speed and steady-state performance [27]. At time sample n, the jth output signal is expressed as

$$y_j(n)=\sum_{i=1}^{I}\mathbf{w}_{ji}^{\mathrm{T}}(n)\mathbf{x}_i(n). \tag{5}$$

The length of each block for the frequency-domain adaptive algorithm is set to B (usually B is identical to L). In the kth frame, the ith reference signal block is $\mathbf{x}_i(k)=\left[x_i(kB-B+1),\ldots,x_i(kB)\right]^T$ and the jth output signal block is $\mathbf{y}_j(k)=\left[y_j(kB-B+1),\ldots,y_j(kB)\right]^T$. The disturbance and error signal blocks at the mth microphone are $\mathbf{d}_{\mathrm{m}}(k)=\left[d_{\mathrm{m}}(kB-B+1),\ldots,d_{\mathrm{m}}(kB)\right]^T$ and $\mathbf{e}_{\mathrm{m}}(k)=\left[e_{\mathrm{m}}(kB-B+1),\ldots,e_{\mathrm{m}}(kB)\right]^T$, respectively. According to the overlap-save method [45], two blocks of the reference signal are concatenated and transformed into the frequency-domain as

$$\mathbf{X}_i(k)=\mathbf{F}\left[\mathbf{x}_i^T(k-1),\mathbf{x}_i^T(k)\right]^T, \tag{6}$$

where $\mathbf{F}$ denotes the discrete Fourier transform (DFT) matrix, whose (p,q)th element is $[\mathbf{F}]_{p,q}=\exp\left(-\mathrm{j}\frac{2\pi pq}{2L}\right)$ with $\mathrm{j}=\sqrt{-1}$, $0\le p\le 2B-1$, and $0\le q\le 2B-1$ [22,24]. The Fourier transform of the error signal vector is given by

$$\mathbf{E}_m(k)=\mathbf{F}\left[\mathbf{0}_{1\times B},\mathbf{e}_{\mathrm{m}}(k)\right]^T, \tag{7}$$

where $\mathbf{0}_{1\times B}$ is the zero vector of B taps. The mth error signal block $\mathbf{e}_{\mathrm{m}}(k)$ is derived as

$$\mathbf{e}_{\mathrm{m}}(k)=\mathbf{d}_{\mathrm{m}}(k)+\sum_{j=1}^{J}\mathbf{G}^{01}\mathbf{F}^{-1}\left[\hat{\mathbf{S}}_{\mathrm{m}j}\odot\mathbf{Y}_j(k)\right], \tag{8}$$

where $\mathbf{F}^{-1}$ denotes the inverse DFT matrix; $\mathbf{G}^{01}=\mathbf{F}\begin{bmatrix}\mathbf{0}_{1\times B} & \mathbf{0}_{1\times B}\\ \mathbf{0}_{1\times B} & \mathbf{1}_{1\times B}\end{bmatrix}\mathbf{F}^{-1}$ is a windowing matrix to preserve the last B elements of the corresponding time-domain vector [22,24]; $\odot$ represents the Hadamard product; $\hat{\mathbf{S}}_{\mathrm{m}j}=\mathbf{F}\left[\hat{\mathbf{s}}_{\mathrm{m}j},\mathbf{0}_{1\times B}\right]^T$ represents the frequency-domain vector of the estimated secondary path. The frequency-domain filtered-reference signal vector is denoted as

$$\mathbf{R}_{mji}(k)=\mathbf{F}\left[\mathbf{r}_{mji}^T(k-1),\mathbf{r}_{mji}^T(k)\right]^T, \tag{9}$$

Table 2
VSS-NFBFxLMS algorithm.

Operation	Multiplications	Additions
1. Compute the frequency-domain reference signal vector using Eq. (6).	$2BI\log_2 2B$	$2BI\log_2 2B$
2. Compute the frequency-domain error signal vector using Eq. (7).	$2BM\log_2 2B$	$2BM\log_2 2B$
3. Compute the frequency-domain filtered-reference signal vector using Eq. (9).	$2BIJM\log_2 2B$	$2BIJM\log_2 2B$
4. Update $\Lambda_i(k)$ using Eq. (11).	$2BI+2I$	$2BI$
5. Update the step size using Eqs. (12)–(14).	$(B+2)(I+M)$	$B(I+M)+2IM$
6. Update the weight vector using Eq. (10).	$6BIJ(M+\log_2 2B)$	$2BIJ(M+3\log_2 2B)+IJL$

where the filtered-reference signal block is $\mathbf{r}_{mji}(k) = \left[r_{mji}(kB-B+1), \dots, r_{mji}(kB)\right]^T$. The update equation is given by [6,22,24]

$$\begin{aligned}\mathbf{w}_{ji}(k+1) &= \mathbf{w}_{ji}(k) - \mu_{mi}(k)\sum_{m=1}^{M}\mathbf{F}^{-1}\left[\mathbf{G}^{10}\Lambda_i^{-1}(k)\mathbf{R}^*_{mji}(k)\odot\mathbf{E}_m(k)\right] \\ &= \mathbf{w}_{ji}(k) - \mu_{mi}(k)\triangle\mathbf{w}'_{ji}(k),\end{aligned} \tag{10}$$

where the constraint matrix $\mathbf{G}^{10} = \mathbf{F}\begin{bmatrix}\mathbf{1}_{1\times B} & \mathbf{0}_{1\times B}\\ \mathbf{0}_{1\times B} & \mathbf{0}_{1\times B}\end{bmatrix}\mathbf{F}^{-1}$ forces the last B time-domain elements to zero, the superscript $(\cdot)^*$ denotes the complex conjugate operator, $\Lambda_i(k)$ denotes the normalisation of the estimated energy of the ith reference signal vector and is given by

$$\Lambda_i(k) = \beta_1\Lambda_i(k-1) + (1-\beta_1)\mathbf{X}_i(k)\odot\mathbf{X}_i(k), \tag{11}$$

where β_1 is the forgetting factor within the range of $[0.9,\ 1)$. The conventional NFBFxLMS algorithm typically sets the step size to a constant as $\tilde{\mu}$, resulting in compromise between steady-state performance and convergence speed with weak robustness under some conditions [26]. In Eq. (10), $\mu_{mi}(k)$ denotes the adjustable step size. Following the step size adjustment strategy proposed by Jiang et al. [27], a robust adaptive step size function is thus used as

$$\mu_{mi}(k) = \begin{cases}\dfrac{\tilde{\mu}}{\sqrt{P_{\mathbf{x}_i}(k)+P_{\mathbf{e}_m}(k)+\eta}} & \mu_{mi}(k) < \mu_{\max}\\ \mu_{\max} & \mu_{mi}(k) \geqslant \mu_{\max}\end{cases}, \tag{12}$$

where η is a regularisation factor and $\mu_{\max}$ is a constraint on the upper step size; $P_{\mathbf{x}_i}(k)$ and $P_{\mathbf{e}_m}(k)$ denote the estimated energy of the ith reference signal block and the mth error signal block, respectively, and are expressed as

$$P_{\mathbf{x}_i}(k) = \beta_2 P_{\mathbf{x}_i}(k-1) + (1-\beta_2)\left\|\mathbf{x}_i(k)\right\|_2^2, \tag{13}$$

$$P_{\mathbf{e}_m}(k) = \beta_3 P_{\mathbf{e}_m}(k-1) + (1-\beta_3)\left\|\mathbf{e}_{\mathrm{m}}(k)\right\|_2^2, \tag{14}$$

where β_2 and β_3 are forgetting factors within the range of $[0.9,\ 1)$; $\|\cdot\|$ denotes the Euclidean norm. This step size adjustment strategy simultaneously considers the possible fluctuations in the reference and error signals. Thus, this strategy potentially improves the robustness of the adaptive algorithm with respect to the impact of noise generated by a car driving over ditches and the effects of talking passengers.

Table 2 summarises the proposed VSS-NFBFxLMS algorithm, and the total number of multiplications and additions required per iteration. In addition, the VSS-NFBFxLMS algorithm can also be adopted in the tuning phase of APFM to further improve the NR performance. Moreover, when selecting a fixed filter with the coefficient set to $\mathbf{0}_{1\times L}$ as the initial value of the VSS-NFBFxLMS algorithm (i.e., $\mathbf{w}_{ji}(n) = \mathbf{w}_{Q+2,ji}(n)$), additional initialisations are required as

$$\triangle\mathbf{w}'_{ji}(k) = \beta_1\triangle\mathbf{w}'_{ji}(k-1),\ \Lambda_i(k)=1,\ P_{\mathbf{x}_i}(k)=0,\ P_{\mathbf{e}_m}(k)=0. \tag{15}$$

3. Simulation with measured transfer functions

To evaluate the effectiveness of the proposed APFM compared to the commonly used ANC algorithms, a series of numerical simulations are performed at a sampling rate of 8 kHz. As shown in Fig. 3, all impulse responses of the used primary and secondary paths in this multi-channel ANC headrest system are measured and modelled as finite impulse response (FIR) filters of 256 taps [46–48]. It is assumed that the estimated secondary paths are the same as the actual secondary paths, as presented in A. The length of the control filter was set to 256 taps. The reference signal is obtained with an accelerometer attached to the primary loudspeaker, which plays the noise to satisfy the causality constraint [4]. Thus, the dimensions of this system are $I=1, J=M=2, L=G=256$. Adding zero-mean bandlimited Gaussian white noise (BWGN) to the interference at the error microphone to simulate the measurement noise, with the signal-to-noise ratio (SNR) set at 40 dB.

In the tuning phase, the step size for the APFM to generate the optimal control filters corresponding to all potential noises is set to 0.001 to ensure accurate estimation. During the control phase, the performances of the FxLMS algorithm, NFBFxLMS algorithm, and the proposed APFM in suppressing the varying noises are compared. Using the trial-and-error method, the step sizes for the FxLMS

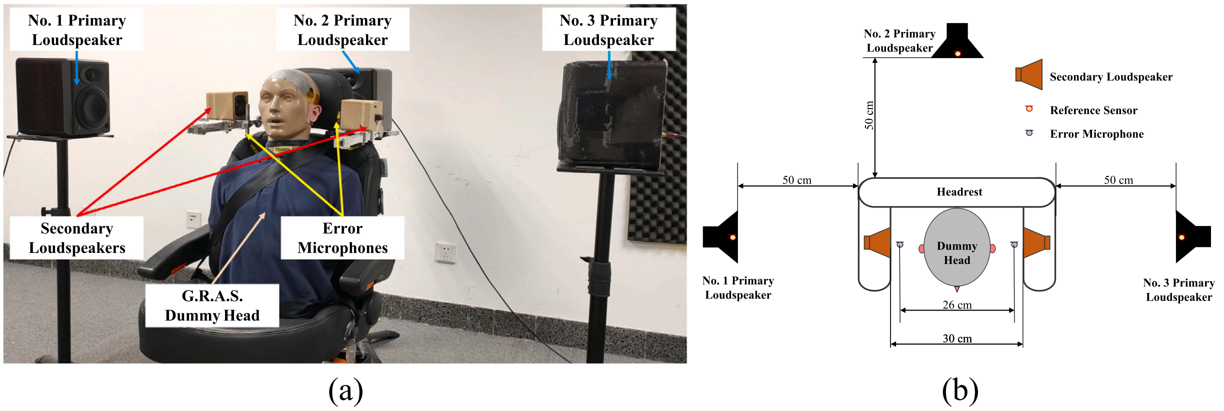

Fig. 3. The configuration of the multi-channel ANC headrest system. (a) Overall setup and (b) schematic diagram.

and NFBFxLMS algorithms are set to 0.01 and 0.1, respectively. The maximum limit of step size for the APFM is set to $\mu_{\max} = 2$. Moreover, the block length is $B = 256$ for the NFBFxLMS algorithm and APFM. Forgetting factors are chosen as $\beta_1 = \beta_2 = \beta_3 = 0.95$.

Considering that the right and left secondary paths are significantly similar, only the results of the right channel are presented in the simulations to avoid confusion due to excessive curves. Average noise reduction (ANR) is used as a metric for intuitive evaluation of the convergence speed and steady-state performance [19], and is defined as

$$\mathrm{ANR}_m = 20\log_{10}\frac{R_{e_m}(n)}{R_{d_m}(n)}. \tag{16}$$

where $R_{e_m}(n) = \alpha R_{e_m}(n-1) + (1-\alpha)\left|e_m(n)\right|$ and $R_{d_m}(n) = \alpha R_{d_m}(n-1) + (1-\alpha)\left|d_m(n)\right|$ are the recursive estimates of the mth error signal and the mth disturbance signal with initial conditions $R_{e_m}(0) = 0,\ R_{d_m}(0) = 0$, and $\alpha = 0.9999$ [19].

3.1. Case 1: BWGN input

In Case 1, the primary loudspeaker generated noise in a sequence of 1→2→3→1→2→3 with each period lasting 15 s, to investigate how the proposed APFM would track changes in the stationary disturbance. At each time point, only one primary loudspeaker emitted BWGN. The first group of BWGNs used to drive the three primary loudspeakers exhibited frequency band ranges of 100–500, 200–600, and 300–700 Hz. The frequency band ranges of the second group of BWGNs were 300–700, 400–800, and 500–900 Hz, individually. Only the first group of BWGNs was used to train the corresponding optimal control filters. Thus, there are $Q = 3$ types of pre-trained primary noises for the proposed APFM.

Fig. 4 shows the ANR curves obtained as an average of 50 independent trials for different ANC algorithms. The corresponding amplitude spectrograms for the different algorithms are shown in Fig. 5. It is observed that the NFBFxLMS algorithm outperforms the FxLMS algorithm in terms of convergence and steady-state performance [6]. The proposed APFM achieves faster convergence speed and lower steady-state error than the NFBFxLMS and FxLMS algorithms. And it is observed that the error signal jitter of the APFM is significantly smaller than that of the other algorithms when the noise changes suddenly. In particular, for non-pre-trained broadband disturbances during the 45–90 s period, the APFM still exhibits the best NR performance. By combining with the VSS-NFBFxLMS algorithm, the APFM benefits from being able to generate the corresponding convergent control filters within 2 s ($\leqslant$2 dB jitter). For non-pre-trained primary noise, the adopted VSS strategy adjusts a large step size to achieve fast convergence when there is a mutation in the noise and turns down the step size in the steady-state stage to obtain a small residual error. Thus, it is confirmed that the APFM is robust to quickly respond to and track the varying stationary broadband noise, and is superior to the two commonly used adaptive algorithms.

3.2. Case 2: BWGN input with speech interference

ANC systems for attenuating vehicle interior road noise are susceptible to the influence of passengers' speech [9,49]. Considering that the reference signals are obtained from non-acoustic sensors, there is poor correlation between speech signals and reference signals [6,13,15]. Hence, the effect of speech is little when noise is suppressed with a fixed control filter [9]. However, speech can affect the received error signals, making it difficult for conventional adaptive algorithms to converge [49]. To validate the robustness of the proposed APFM against speech interference, this case evaluates the NR performance of different algorithms for various SNR levels of speech interference signal.

In Case 2, the primary loudspeaker produced noise in the order of 1→2→3, with a duration of 15 s for each period. The speech signal used as interference was obtained from the Librispeech dataset [50]. The waveform and spectrogram of the speech interference are shown in Fig. 6. The speech interference was only added to the error signal and the corresponding SNR was set to −6, 0, and

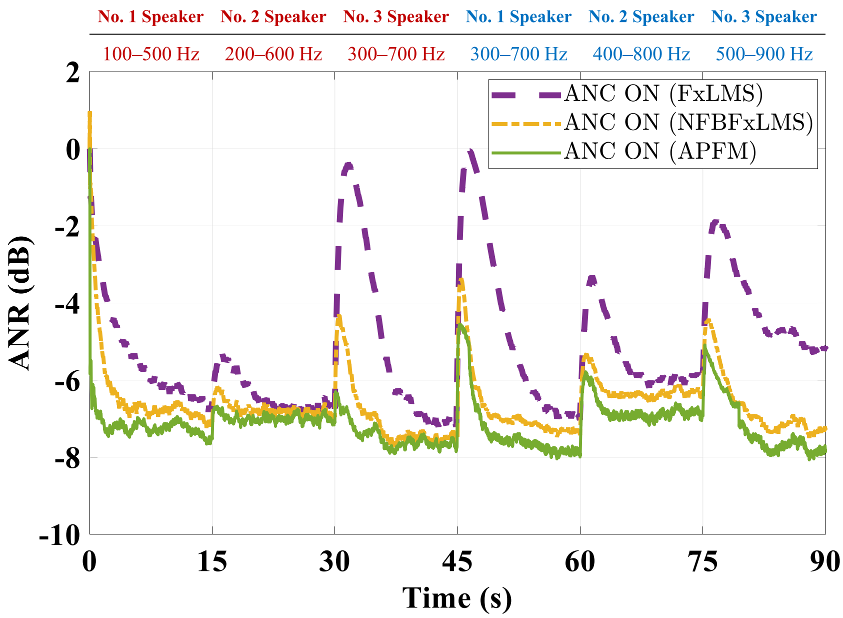

Fig. 4. ANR curves for different algorithms in Case 1.

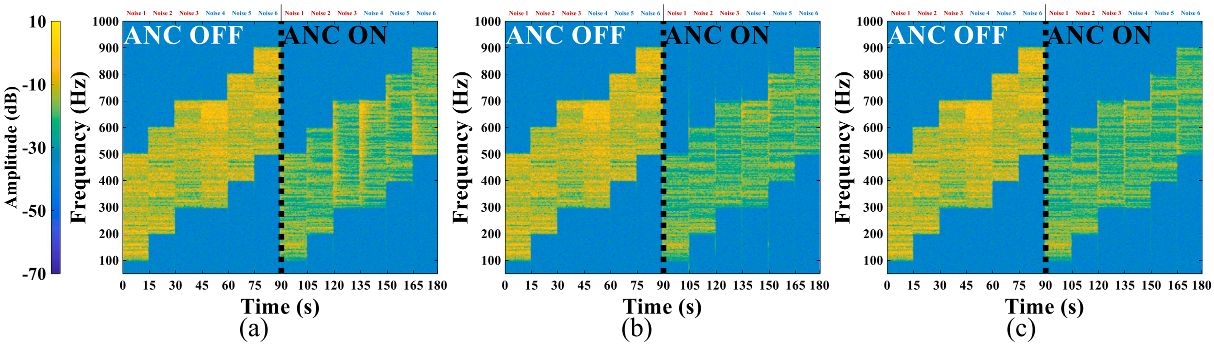

Fig. 5. Amplitude spectrogram showing the NR levels of the different algorithms for BWGN inputs. (a) FxLMS, (b) NFBFxLMS, and (c) APFM.

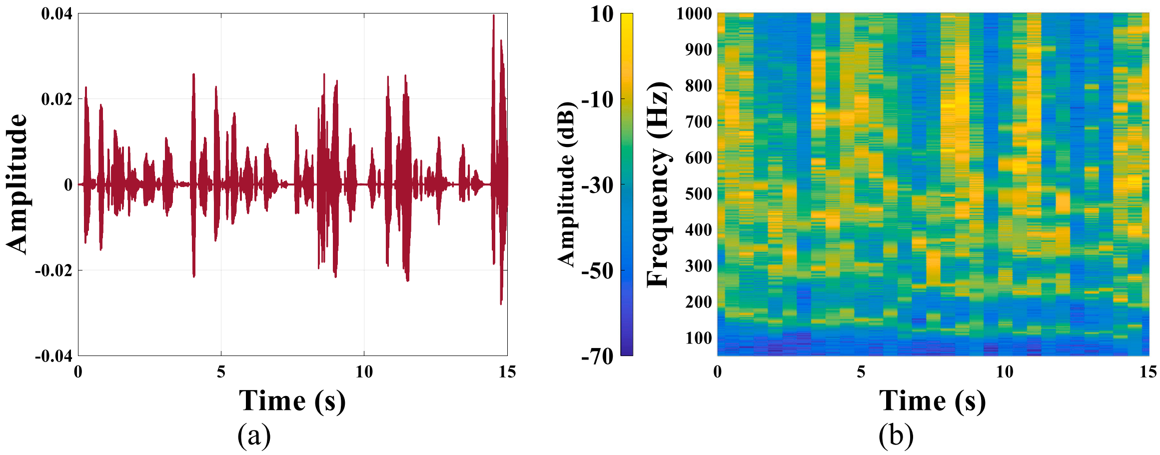

Fig. 6. (a) Waveform and (b) spectrogram of the adopted speech interference.

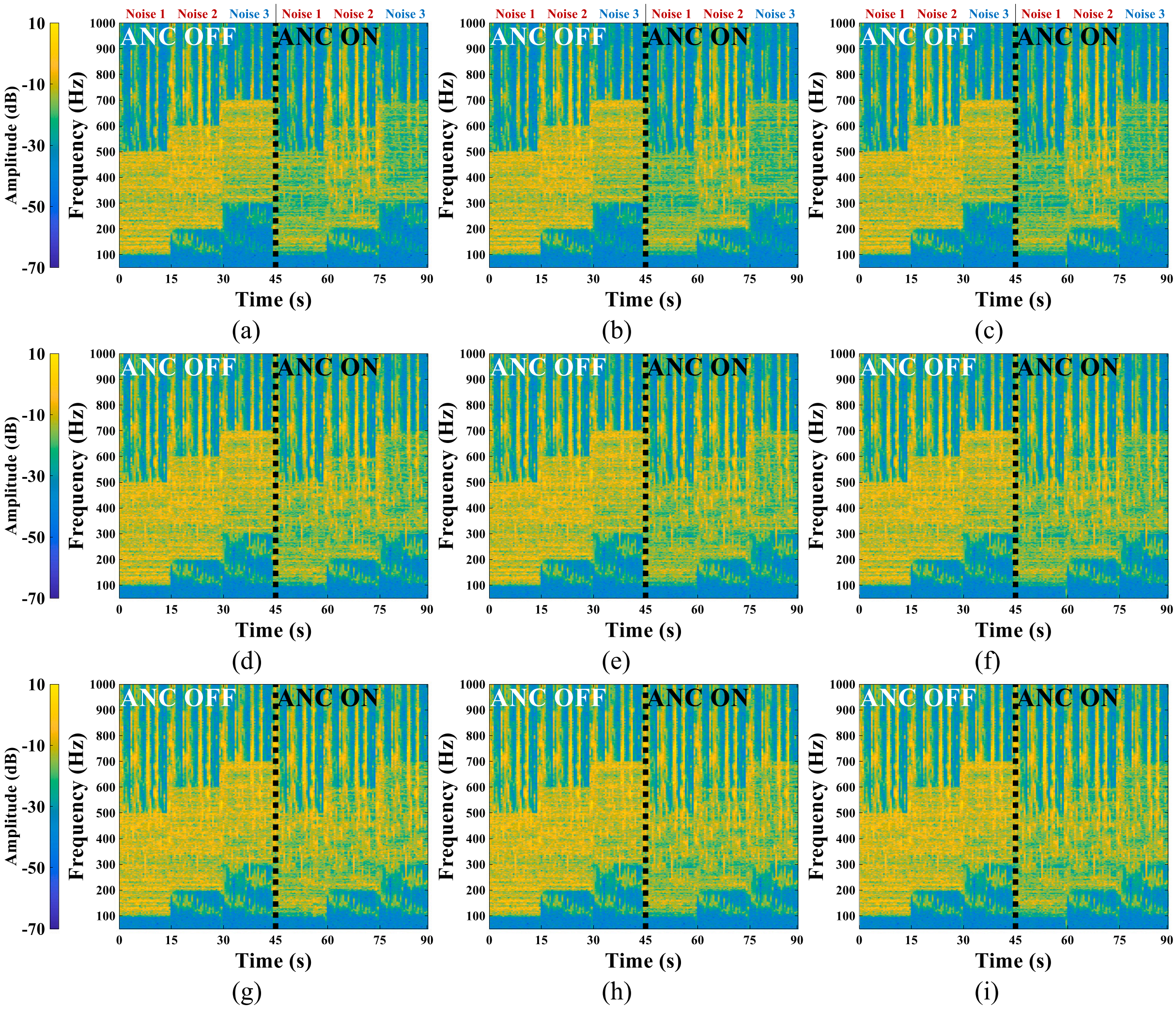

Fig. 7. Amplitude spectrogram showing the NR levels of the different algorithms for BWGN inputs with speech interference. (a) FxLMS, −6 dB, (b) NFBFxLMS, −6 dB, (c) APFM, −6 dB, (d) FxLMS, 0 dB, (e) NFBFxLMS, 0 dB, (f) APFM, 0 dB, (g) FxLMS, 3 dB, (h) NFBFxLMS, 3 dB, and (i) APFM, 3 dB.

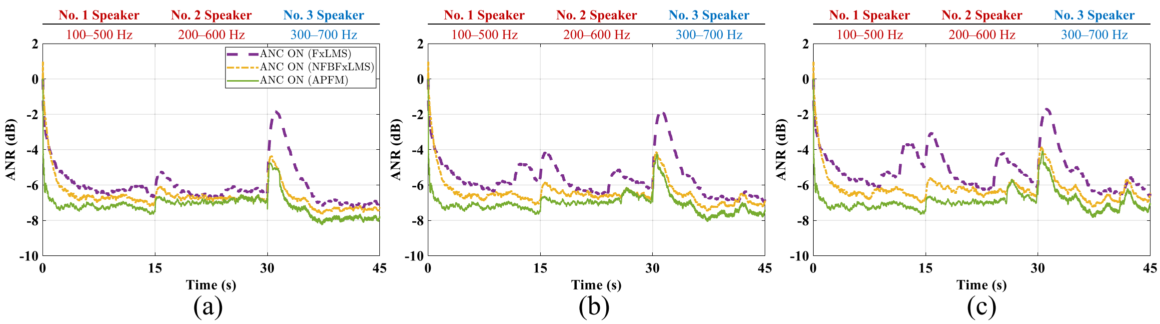

Fig. 8. ANR curves for different algorithms with speech interference ignored in Case 2. (a) −6 dB, (b) 0 dB, and (c) 3 dB.

3 dB, respectively. A higher value of SNR indicates that the corresponding speech interference is more intense in this case. The band ranges of the BWGNs driving the three primary loudspeakers were 100–500, 200–600, and 300–700 Hz. Only the first two types of noise were used to train the corresponding optimal control filters for the APFM. Correspondingly, there are $Q = 2$ types of pre-trained primary disturbances.

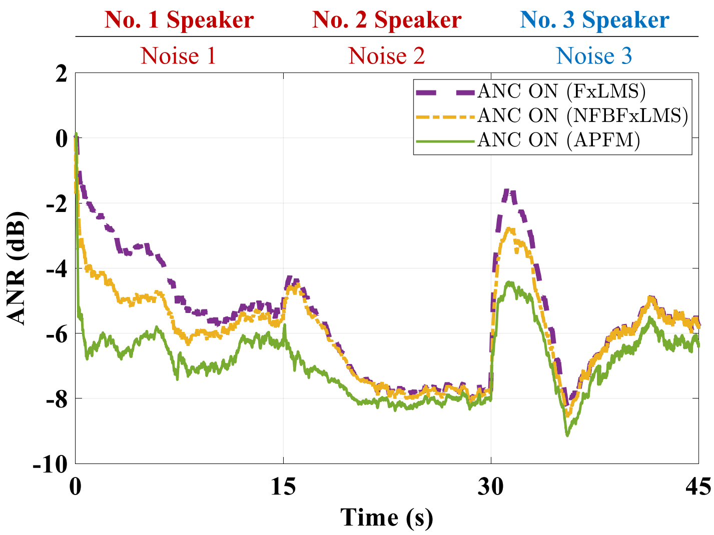

Fig. 9. ANR curves for different algorithms in Case 3.

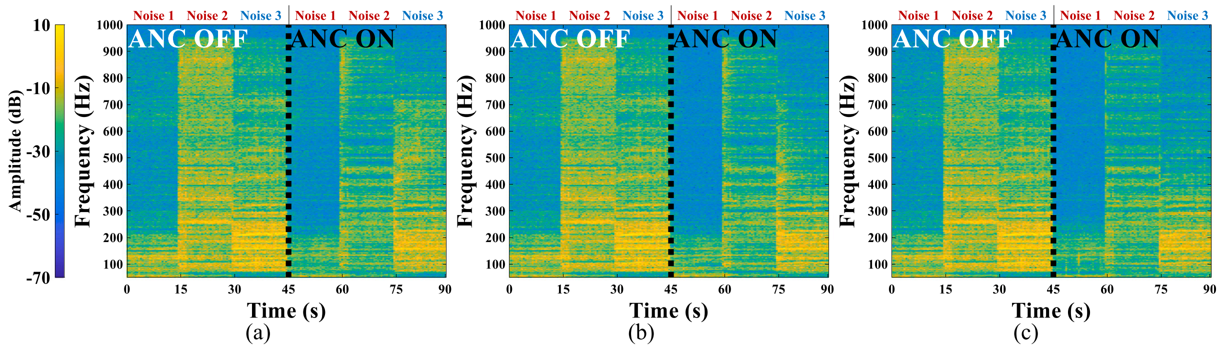

Fig. 10. Amplitude spectrogram illustrating the NR levels of the different algorithms for non-stationary practical noise inputs. (a) FxLMS, (b) NFBFxLMS, and (c) APFM.

The amplitude spectrogram of the error signal, obtained as an average of 50 independent trials for all three ANC algorithms, is shown in Fig. 7. Since speech cannot be suppressed, the level of NR is reduced to varying degrees. Ignoring speech interference, the corresponding ANR curves for the different algorithms are presented in Fig. 8. It is observed that the APFM achieves the fastest convergence speed and the lowest steady-state error when dealing with broadband noise and speech interference with different SNRs, regardless of whether the noise is pre-trained. Moreover, as the energy of the speech interference increases, the ANR curve of the APFM is the lowest affected by speech interference and exhibits the least misalignment at steady-state. This is due to the fact that the proposed APFM is based on the selection mechanism to select suitable control filters and adjusting frequency-domain block adaptive filters using the VSS strategy. During the transition period from one primary noise to another, the residual error signal of the APFM fluctuates the least. Even though there is speech interference, the transient performance of APFM remains significantly superior to that of the other two algorithms when there are sudden noise variations. Therefore, it is confirmed that the APFM is robust in suppressing stationary broadband noise with speech interference. At the same time, the NFBFxLMS algorithm shows a faster convergence speed than the FxLMS algorithm. This is probably because the principal components of the speech interference are rarely in the same frequency bands as those of BWGNs [2,50].

3.3. Case 3: non-stationary practical noise input

In practice, vehicle interior road noise is generally non-stationary [2,7]. To further evaluate the performance of the proposed APFM in attenuating non-stationary practical noise, car interior road noise, aircraft cabin noise, and military vehicle road noise from the NOISEX-92 dataset were used individually to drive the three primary loudspeakers [51]. The rule for the primary loudspeakers generating noise was identical to that in Case 2. Both car interior road noise and aircraft cabin noise were used to train the corresponding optimal control filters for the proposed APFM. It means that $Q = 2$ kinds of primary noises are trained during the tuning stage, and thus $Q+2$ types of parallel filters are adopted in the control phase of the APFM. Fig. 9 depicts the ANR curves for

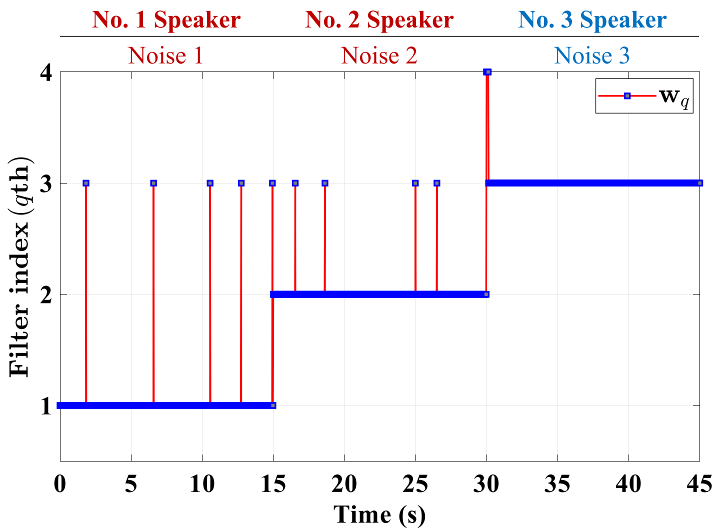

Fig. 11. The index of selected control filter over time by the APFM with $Q = 2$ for attenuating non-stationary practical noise inputs.

the error signals obtained as an average of 50 independent trials for all three ANC algorithms. During the first 30 s, the APFM still exhibits a significant advantage over conventional FxLMS and NFBFxLMS algorithms, as shown by the amplitude spectrogram in Fig. 10. For non-pre-trained military vehicle road noise, the proposed APFM achieves the best convergence speed and steady-state performance by combining the adaptive frequency-domain block algorithm with the VSS method. When there is a sudden change in the primary noise, the error signal jitter of the APFM is minimal due to the selection mechanism and VSS strategy. Moreover, Fig. 11 illustrates the index of the selected control filter over time by the APFM during a trial. From these results, it is clear that the proposed APFM is flexible in selecting the appropriate control filter to attenuate varying noise [34]. Although the pre-trained fixed control filters are ineffective in suppressing untrained military vehicle road noise, the APFM achieves superior NR performance over the FxLMS and NFBFxLMS algorithms by using the VSS-NFBFxLMS algorithm. Thus, it is demonstrated that the proposed algorithm provides excellent suppression of non-stationary practical noise.

From the above simulations, it is evident that the proposed APFM can track and respond quickly to varying broadband noises of different frequency bands and is robust to speech interference. In addition, the APFM benefits from the combination of the delayless VSS-NFBFxLMS algorithm, and achieves superior NR performance in the presence of broadband disturbances regardless of whether the noise is trained in advance. It is recommended to pre-train as many fixed optimal control filters as possible according to potential primary noises in specific practical applications, as long as the arithmetic requirements are met [44].

4. Experimental validation

To comprehensively evaluate the effectiveness of the APFM for road noise cancellation inside vehicles, a multi-channel ANC headrest system with a dummy head (G.R.A.S., KEMAR head & torso) was installed in the right rear seat of an electric vehicle, as shown in Fig. 12. In this ANC headrest system, the distance between the secondary loudspeakers and the error microphones in each pair was 3 cm apart, and the distance between the error microphone and the dummy head was approximately 5 cm. The distance between the error microphones on either side was around 21 cm. The signals measured at the error microphones before and after ANC were used to evaluate the amount of NR. The secondary paths were estimated as 256-tap FIR filters using bandlimited white noise <1 kHz and an adaptive identification algorithm based on the normalised least mean square algorithm [52]. Fig. 13 depicts the impulse responses of the estimated secondary paths in detail. The reference signals were simultaneously acquired from six accelerometers mounted around the subframe mounts and trailing arms. The positions of these sensors were determined by coherence analysis and the effective independence method [3,14]. The dimensions of this ANC system were $I = 6$, $J = M = 2$. The length of control filters L, the length of estimated secondary paths G, and the frame rate (block length) B are all set to 256. The cut-off frequencies of both the anti-aliasing and reconstruction filters were 1 kHz. The ANC controller was implemented on a real time multi-core digital signal processor TMS320C6678 with a sampling rate of 5 kHz.

Based on common driving conditions, the electric vehicle was successively tested on rough and smooth surfaces. The experiments were conducted at three different speeds of 50, 60, and 70 km/h [6,9]. The amplitude spectrum of the disturbance in the right and left channels under various driving conditions is presented in Appendix B. It is clear that even if the speed varies by only 10 km/h, the corresponding road noise inside the vehicle differs substantially. Considering that the speed is generally ever changing in real-life condition, the duration of each speed was approximately 10 s. Additionally, road noises are usually non-stationary [2,7]. It is difficult to determine whether control filters have converged by examining the norm of coefficients for control filters [43]. Hence, this section relies on the method adopted by Buck and Sachau [47] to assess the convergence of the algorithms. As long as over

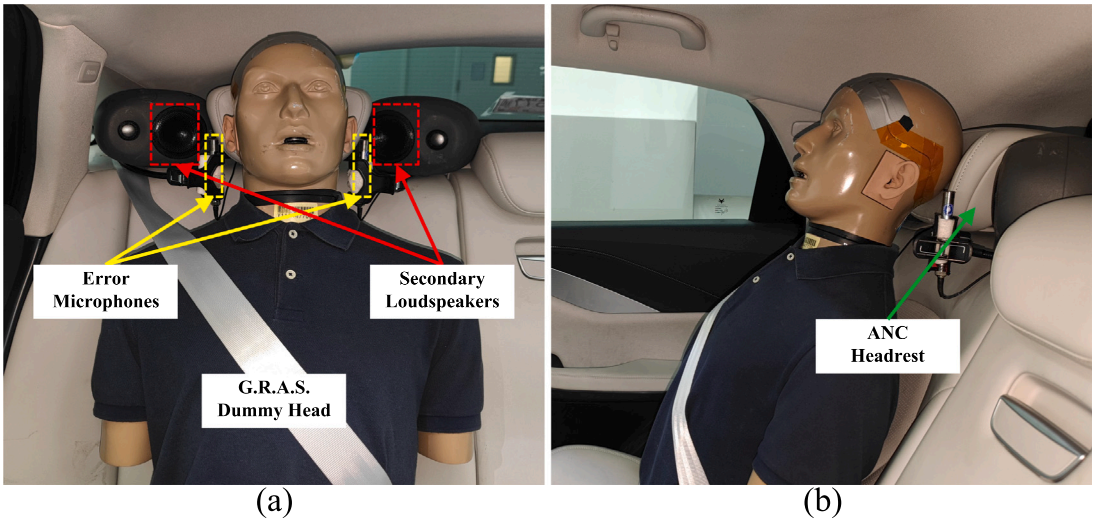

Fig. 12. Experimental configuration of the ANC headrest system in an electric vehicle. (a) Front view and (b) left view.

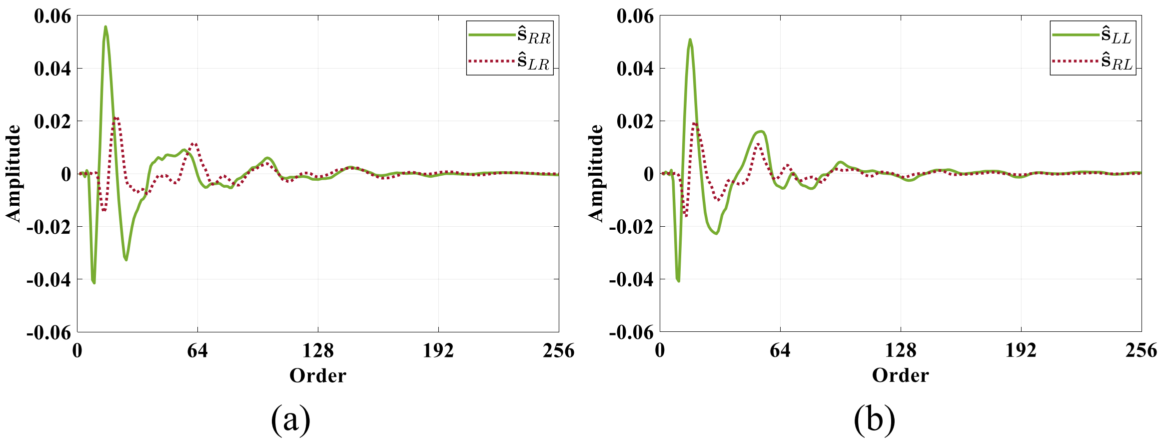

Fig. 13. Impulse responses of the estimated secondary paths $\hat{s}_{mj}$ of the (a) right and (b) left secondary loudspeakers.

3 dB of road noise attenuation is achieved and the jitter in the NR amount is less than 2 dB before the driving condition changes, control filters are considered to have converged. For comparison, the conventional FxLMS and delayless NFBFxLMS algorithms were also evaluated, with the used step sizes μ set to 0.001 and 0.05, respectively. The values of the other parameters were the same as for the simulations. For the APFM, the fixed optimal control filters corresponding to driving on the rough surface at speeds of 50, 60, and 70 km/h were pre-trained using the multi-channel FxLMS algorithm through numerical simulation. Therefore, for six separate driving conditions, $Q = 3$ types of primary noises were trained in advance, which meant that $Q+2$ kinds of parallel control filters were used in the control phase of the APFM. The magnitude responses of the pre-trained fixed control filters are displayed in Fig. 14.

Figs. 15 and 16 show the amplitude spectrum of the A-weighted sound pressure level (SPL) measured at the error microphones under all driving conditions before and after ANC. The amplitude spectrogram of the measured error signals for all three ANC algorithms is displayed in Fig. 17, and Fig. 18 depicts the index of the selected control filters over time by the APFM during a trial. For the pre-trained road noises, the selected control filters were matched with corresponding pre-filters for the transient region when the road noise was suddenly changed. And there was occasional adaptive adjustment of the control filter to rapidly attenuate non-stationary disturbance. As for the non-pre-trained road noises, the control filters adopted during the transition period from one type of road noise to another were generally selected with the corresponding pre-filters obtained at the same speed. The VSS-NFBFxLMS algorithm was then continuously applied to achieve fast convergence speed and low steady-state errors. And the APFM was found to be significantly superior to the two conventional ANC algorithms in suppressing road noise in the frequency range of 50–500 Hz, regardless of whether the noise was pre-trained. As illustrated by the average NR results of the APFM for all driving conditions in Figs. 15 and 16, the booming noise from 80 to 180 Hz, the tyre cavity resonance noise from 200 to 240 Hz, and the rumbling noise

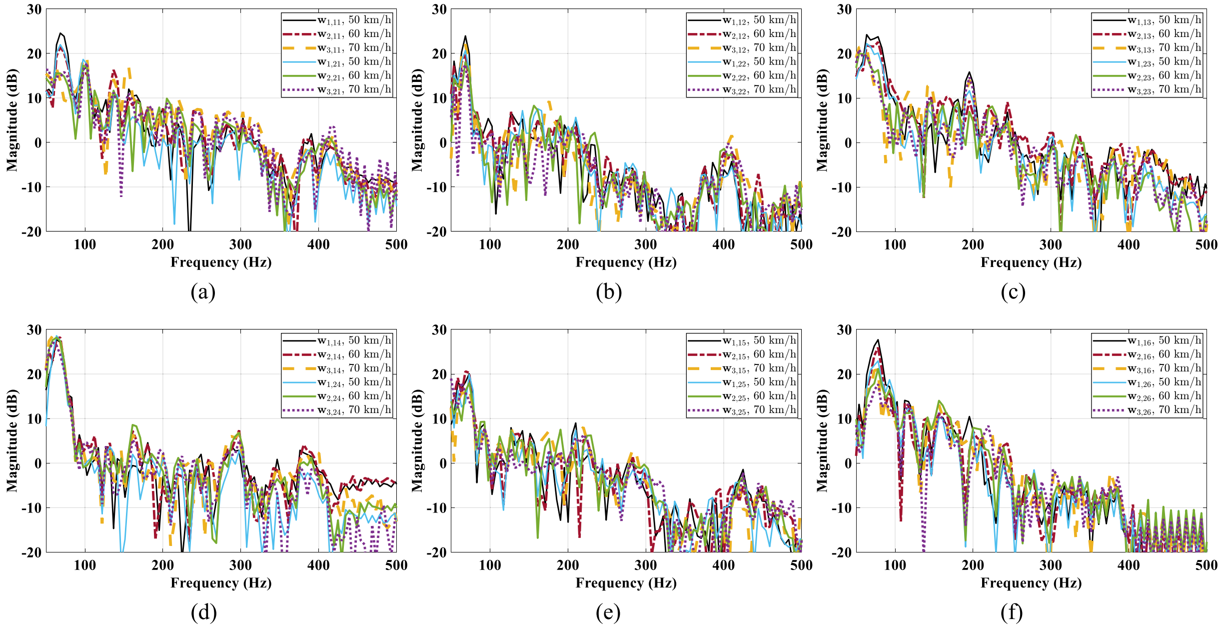

Fig. 14. Magnitude responses of the pre-trained fixed control filters $\mathbf{w}_{q,ji}$ corresponding to the (a) first, (b) second, (c) third, (d) fourth, (e) fifth, and (f) sixth reference signals.

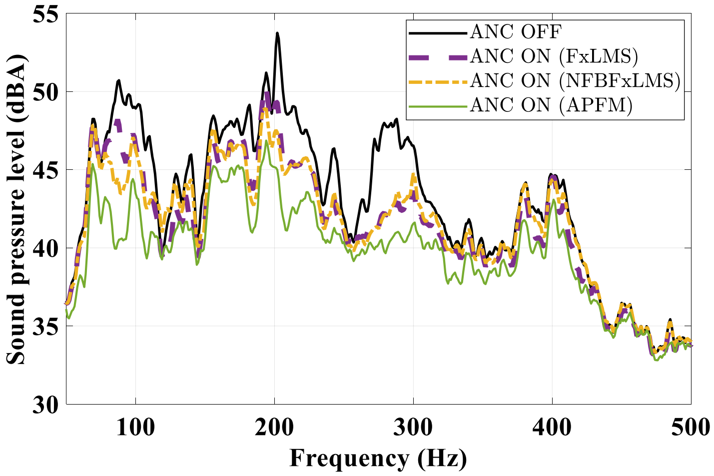

Fig. 15. The measured A-weighted SPL at the right microphone under all driving conditions with different ANC algorithms.

from 250 to 350 Hz were suppressed by more than 3 dBA at the error microphones [15]. And the APFM achieved an attenuation of approximately 10 dBA at the 202 Hz peak. In comparison, the two conventional ANC algorithms exhibited inferior and unstable NR performance when reducing varying primary noise, especially on the rough surface.

Table 3 lists the attenuation of various road noises. At the same speed, the amount of NR attained by the APFM was generally higher on the rough surface than on the smooth surface. The NR reached in the right channel on the rough surface was usually higher than that in the left channel. This is because the SPL of background noise is enhanced in the right channel, allowing the APFM to take full advantage of the potential performance of fast tracking and superior noise cancellation. The road noise generated by driving on the smooth surface tended to stabilise [2]. Thus, the difference in NR performance among the three ANC algorithms was comparatively slight [6]. In detail, the APFM achieved similar NR levels at different speeds on the smooth surface. And it was noticed that the NR attained at the right microphone was less than that achieved at the left microphone at 50 km/h. To investigate the potential reason of this phenomenon, fixed control filters were trained for this driving condition individually, and the corresponding NR amount was investigated. The residual errors after suppression by the fixed control filters were fundamentally identical to that achieved by the APFM. Hence, the secondary paths used may limit the NR attained at the right error microphone under this driving condition [53].

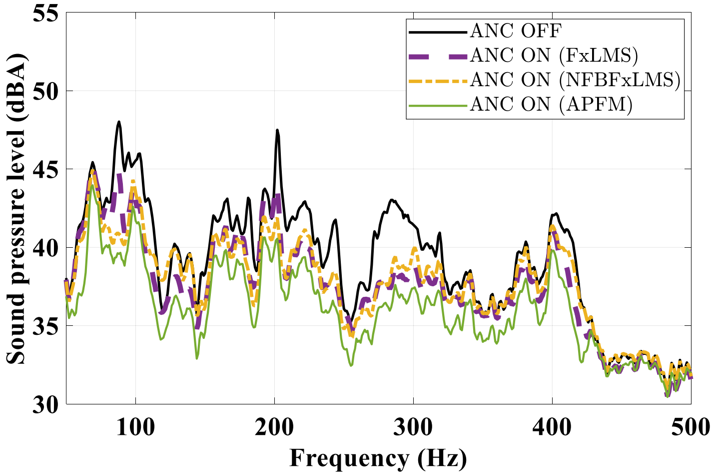

Fig. 16. The measured A-weighted SPL at the left microphone under all driving conditions with different ANC algorithms.

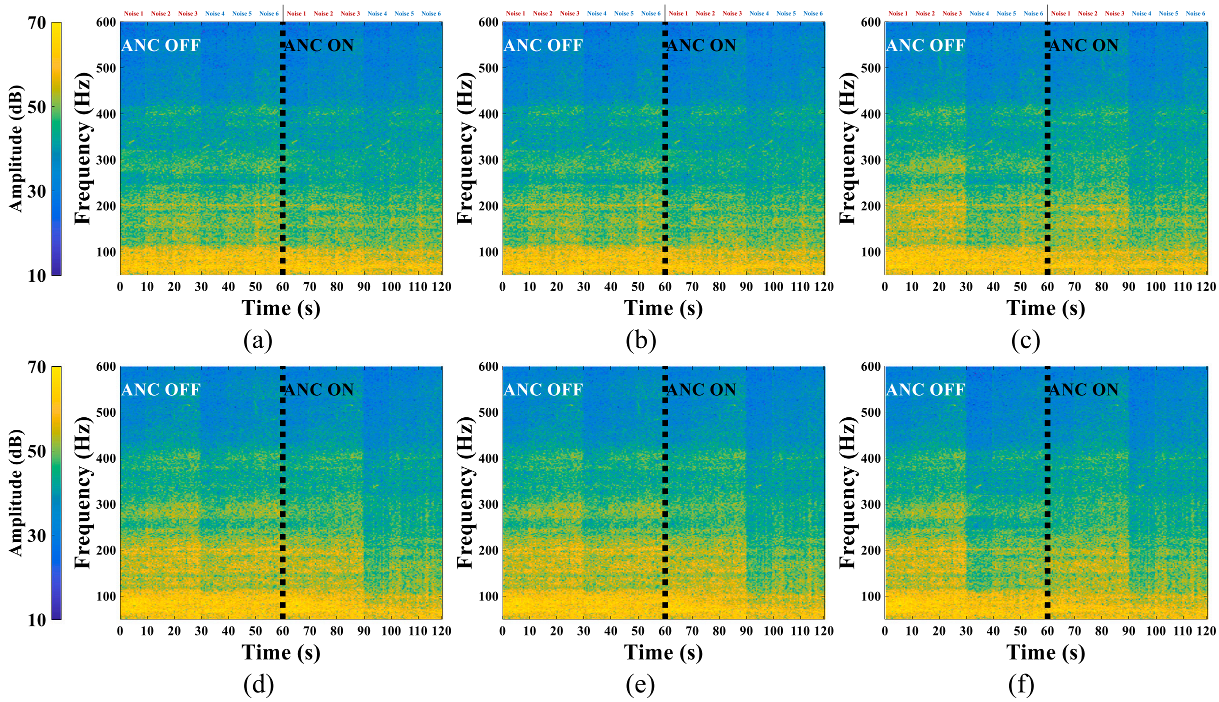

Fig. 17. Amplitude spectrogram showing the NR levels of the different algorithms for varying road noises. (a) FxLMS, right channel, (b) NFBFxLMS, right channel, (c) APFM, right channel, (d) FxLMS, left channel, (e) NFBFxLMS, left channel, and (f) APFM, left channel.

Figs. 19 and 20 compare the ANR curves of the three ANC algorithms over ten independent trials. After switching to different driving conditions for approximately 2 s, the APFM converges (⩽2 dB jitter). As the vehicle speed increases, the interior road noise changes faster and tends to become more non-stationary [2]. Therefore, the NR performance of the APFM tends to degrade. For all configurations considered, it is found that the APFM performs best in terms of convergence speed and steady-state performance. Whether the road noise is pre-trained or not, the APFM exhibits a fast response and superior tracking capability. As driving conditions transition from one to the other, the APFM takes full advantage of the selection mechanism and VSS strategy to obtain noticeable suppression of non-stationary road noise. Besides, the NFBFxLMS algorithm usually converges faster and achieves a lower residual error than the FxLMS algorithm. This is because the time-domain algorithms are more susceptible to the input signal power, resulting in a slow convergence speed and a large steady-state error [6].

Although the distance from the secondary loudspeaker to the error microphone was fixed in the experiments, different head scattering effects may introduce slight variations in the secondary path. To further verify the potential impact of secondary path

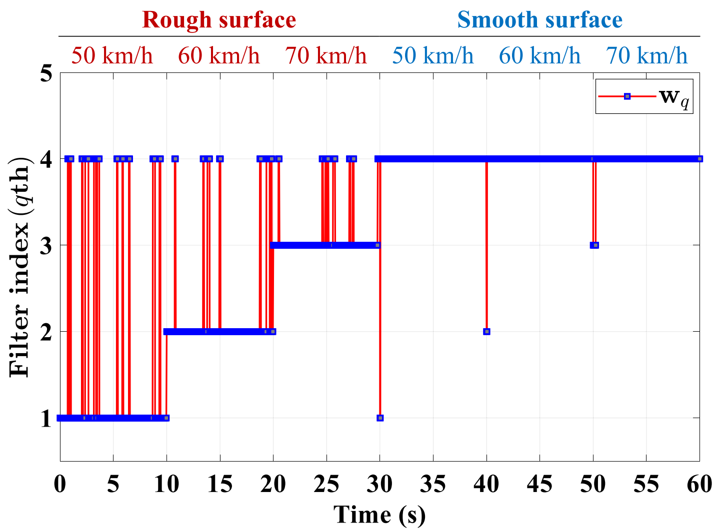

Fig. 18. The index of selected control filter over time by the APFM with $Q = 3$ under various driving conditions.

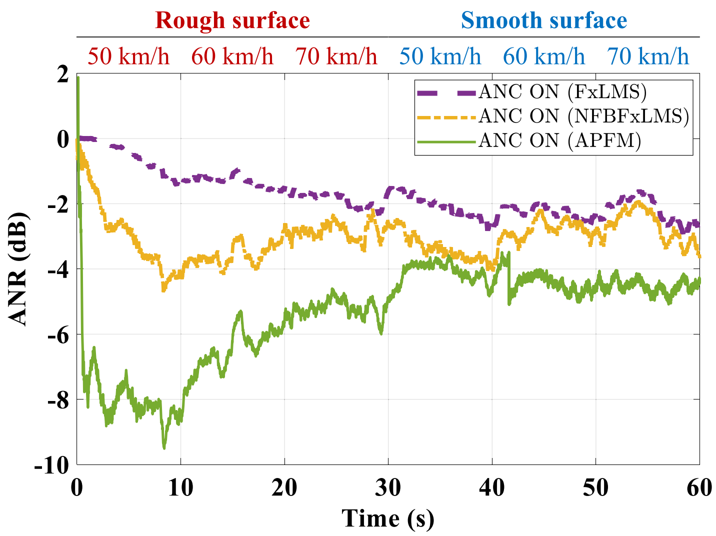

Fig. 19. Performance comparison based on ANR at the right error microphone under different driving conditions.

Table 3
Comparison of attenuation (dBA) obtained by using different ANC algorithms for various driving conditions on the right/left channel.

Driving condition		ANC algorithm		
		FxLMS	NFBFxLMS	APFM
Rough surface	50 km/h	0.9/0.5	2.5/1.6	5.2/4.6
	60 km/h	1.3/0.8	2.3/1.4	4.4/4.1
	70 km/h	1.5/1.1	2.4/1.5	3.8/3.7
Smooth surface	50 km/h	1.8/1.6	2.5/2.4	3.2/3.9
	60 km/h	1.9/1.8	2.3/2.3	3.3/3.2
	70 km/h	1.8/1.7	2.2/1.9	3.4/3.0

perturbations on the APFM, a passenger weighing approximately 77 kg was invited to conduct experiments under the same six driving conditions. The estimated secondary paths used for the APFM were those modelled in the previous experiments. The magnitude and phase responses of these cross-channel secondary paths are presented in Fig. 21. As shown in Fig. 21, with the relative

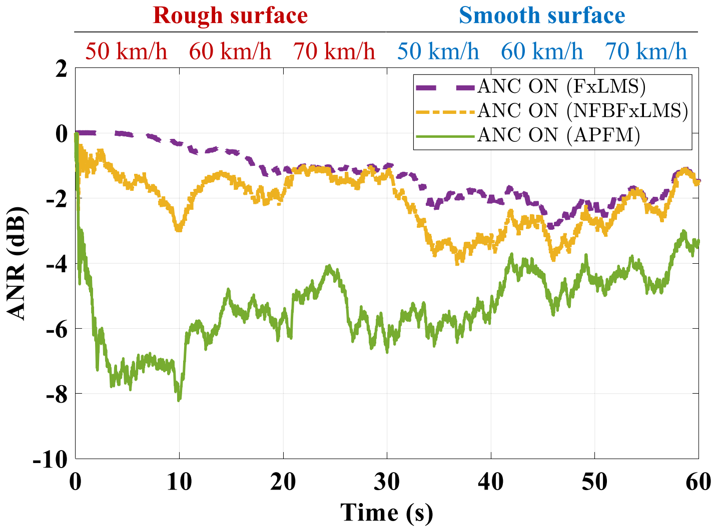

Fig. 20. Performance comparison based on ANR at the left error microphone under different driving conditions.

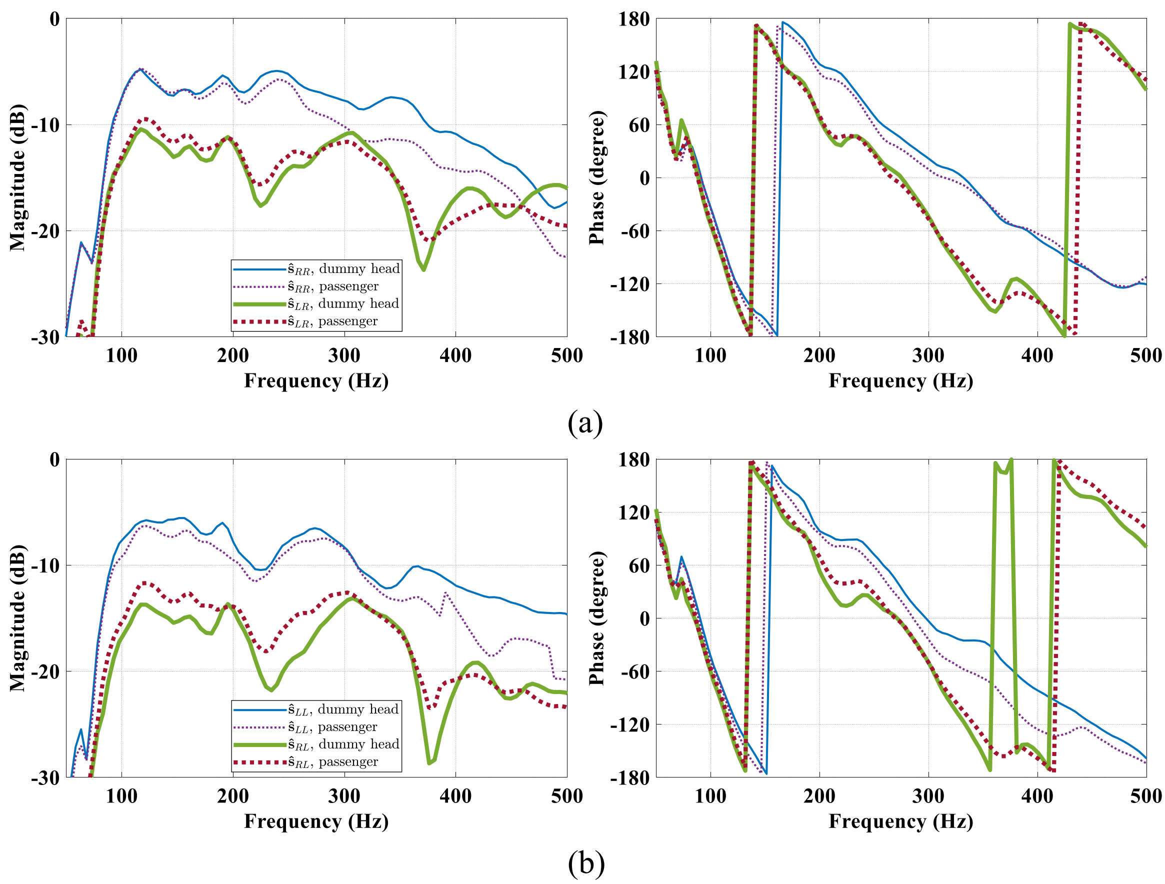

Fig. 21. Magnitude and phase responses of cross-channel secondary paths measured under different head conditions. (a) $\hat{s}_{RR}$, $\hat{s}_{LR}$, and (b) $\hat{s}_{LL}$, $\hat{s}_{RL}$.

position between the secondary loudspeaker and the error microphone fixed, the secondary path was susceptible to perturbations by scattering effects of the different heads [5,54]. Fig. 22 displays the measured A-weighted SPL at the error microphones during the

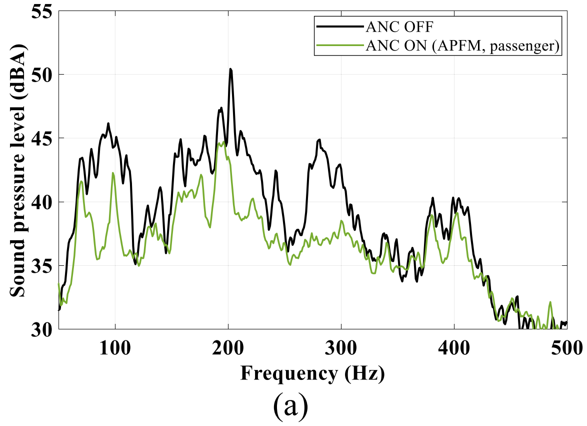

Fig. 22. A-weighted SPL measured at the (a) right and (b) left error microphones when the passenger test drive caused a slight variation in the secondary paths.

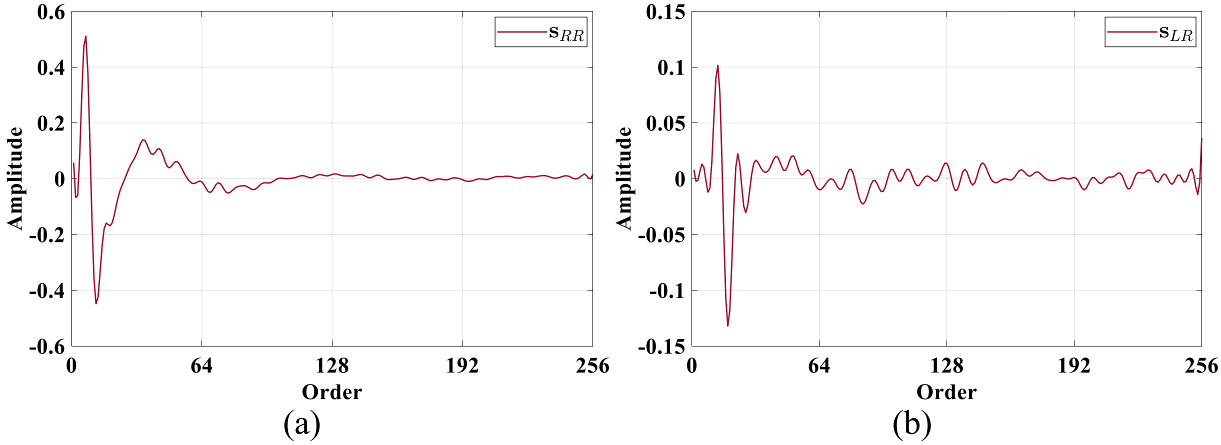

Fig. A.23. Secondary paths from the right secondary source to (a) right and (b) left microphones.

passenger test drive. As observed in Figs. 15, 16, and 22, different head conditions induced secondary path variations, which led to the NR performance of the APFM degradation. Low accuracy of the secondary path estimation tends to mismatch the control filter due to the inferior precision in estimating residual errors [44]. The proposed VSS-NFBFxLMS algorithm partly compensated for the effect of control filter mismatch on the NR performance of the APFM. Road noises below 300 Hz were still significantly attenuated. Booming noise, tyre cavity resonance noise, and rumbling noise, which caused subjective auditory discomfort to the passenger, were effectively suppressed [15]. Moreover, the positive subjective assessment given by the passenger also proves that the varying road noises are significantly attenuated and the APFM is effective for human auditory perception [54]. For future applications of the APFM, introducing online secondary path modelling and multi-sensor fusion to track targets may further alleviate the potential adverse effect of secondary path variations on the APFM [14,55].

Under the above driving conditions, the APFM showed the best performance in suppressing vehicle interior road noise. Compared to the other algorithms, the APFM exhibited superior performance in terms of convergence speed, steady-state noise suppression, and robustness to varying road noise. The experimental results confirm that the APFM provides fast response and promising tracking capability in suppressing changing primary disturbances. These results support the theoretical and simulation analyses presented in the previous sections. Although the positions of all six reference sensors adopted in the real time experiments were painstakingly selected, causality conditions may be unsatisfied under certain driving conditions [13]. And it will provoke the frequency-domain block adaptative algorithm to converge to a biased solution, thus limiting the NR performance of the APFM, especially in suppressing non-pre-trained road noises [23,24]. To fulfil the causality constraints under various driving conditions, the sampling rate and the number of reference signals should be increased as much as possible. Meanwhile, the APFM can be regarded as a potential scheme for various practical applications, such as ANC headphones, ANC casings, and ANC barriers [56–61].

Moreover, the NR performance achieved by different ANC algorithms in the experiment was obtained from two microphones around the ears of the dummy head. Virtual sensing methods are required to compensate for the degradation of NR when physical microphones cannot be placed at the target [13,62,63]. In addition, APFM requires a high degree of the combined adaptive filter

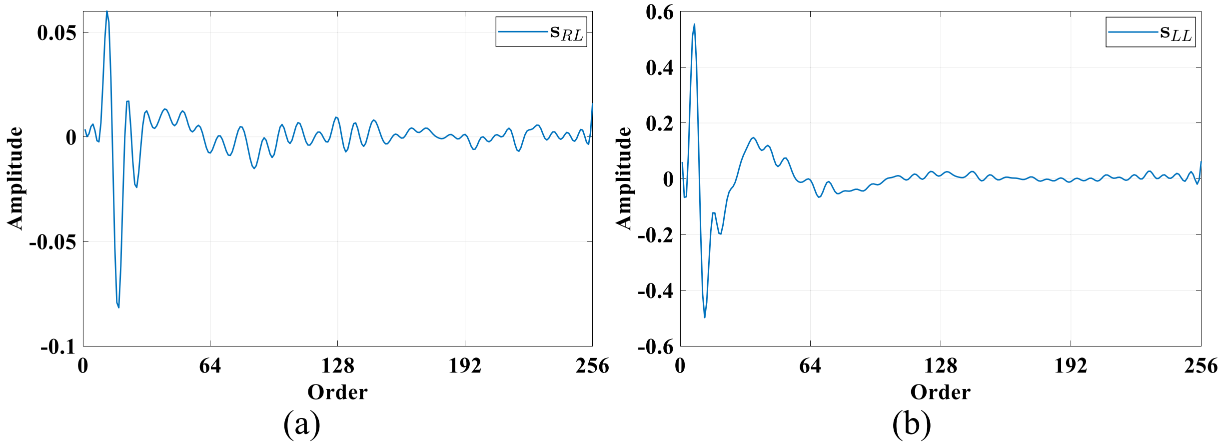

Fig. A.24. Secondary paths from the left secondary source to (a) right and (b) left microphones.

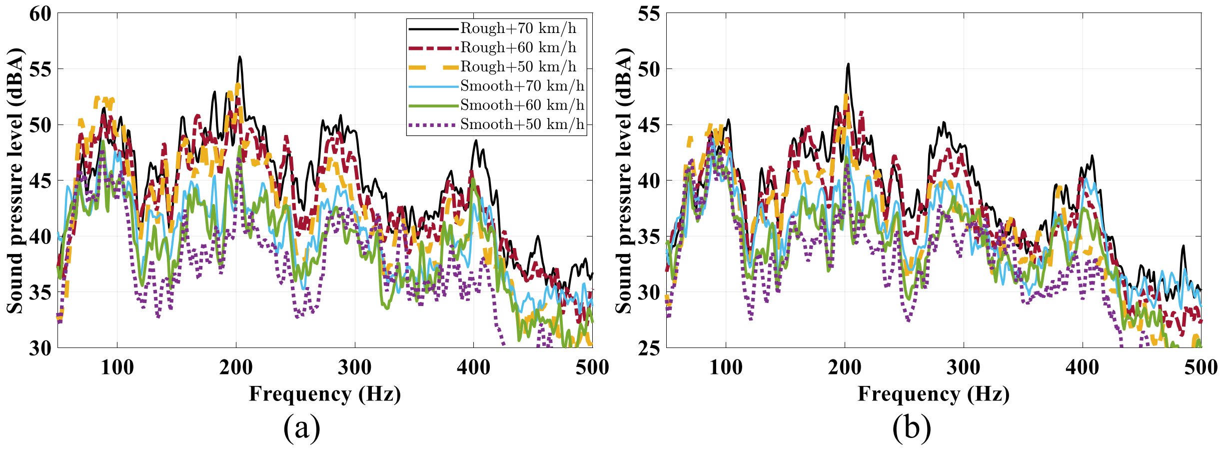

Fig. B.25. A-weighted SPL of vehicle interior road noises measured at (a) right and (b) left microphones under different driving conditions.

algorithm. Delayless frequency-domain partitioned block adaptive algorithms and subband adaptive algorithms can be adopted in ANC systems to reduce the computational cost and increase convergence speed [23,47]. In the future, we will address these issues.

5. Conclusions

To efficiently attenuate varying vehicle interior road noise, this study proposes a robust APFM that achieves fast NR response and low steady-state error. The APFM takes advantage of fixed-filter methods and adaptive filter algorithms for ANC systems. In the tuning phase, a bank of parallel optimal control filters is modelled to account for potential noises that may appear. During the control phase, the most appropriate parallel control filter is first selected at the frame rate based on the selection mechanism to minimise the energy of the estimated error signal. Then, the delayless VSS-NFBFxLMS algorithm is adopted to adaptively update the coefficients of the selected control filter, which improves the performance in suppressing various types of noise with low computational cost. Numerous simulations were performed with measured transfer functions under different disturbance conditions. The increased computational cost of the APFM is tolerable, and its robust NR performance against varying broadband noise compensates for this deficiency. Moreover, real time experiments were conducted on the multi-channel ANC headrest in the electric vehicle under different driving conditions. It is validated that the APFM outperforms the commonly used adaptive ANC algorithms in terms of NR response time, steady-state performance, and robustness in suppressing varying road noise inside the vehicle.

CRediT authorship contribution statement

Lan Yin: Methodology, Investigation, Writing – original draft, Writing – review & editing. **Zeqiang Zhang:** Conceptualization, Methodology, Formal analysis, Writing – original draft. **Ming Wu:** Conceptualization, Methodology, Validation. **Zhiliang Wang:** Investigation, Validation. **Chao Ma:** Investigation, Validation. **Shuang Zhou:** Methodology, Validation. **Jun Yang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Secondary paths

Figs. A.23 and A.24 show the impulse responses of the used secondary paths in simulations.

Appendix B. Measured disturbances

Fig. B.25 shows the A-weighted SPL of the disturbances at the microphones under different driving conditions.

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